SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Exploratory Data Analysis: Statistical Modeling & Data Cleaning

Department: CSE DS	**Course / Subject:** Exploratory Data Analysis (DS401PC)
Semester: Semester 4	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Descriptive statistics, Outlier detection (IQR, Z-Score), Probability distributions, Hypothesis testing (t-test, ANOVA, Chi-Square), and Multivariate analysis.

Unit 1: Data Understanding & Cleaning

Types of variables (Nominal, Ordinal, Interval, Ratio), Handling missing data (MCAR, MAR, MNAR), Imputation techniques, and duplicate identification.

Unit 2: Descriptive & Summary Statistics

Measures of central tendency (Mean, Median, Mode), Measures of dispersion (Variance, Standard Deviation, Interquartile Range IQR), Skewness, and Kurtosis.

Unit 3: Probability Distributions

Discrete distributions (Binomial, Poisson), Continuous distributions (Normal, Standard Normal, Exponential), Central Limit Theorem, and Sampling techniques.

Unit 4: Inferential Statistics & Hypothesis Testing

Null and Alternative hypotheses, Type I and Type II errors, p-values, One-sample and two-sample t-tests, Paired t-test, One-way ANOVA, and Chi-Square tests of independence.

Unit 5: Bivariate & Multivariate Exploration

Covariance, Pearson's correlation coefficient, Spearman rank correlation, Collinearity detection (VIF), Factor Analysis, and Principal Component Analysis.